AEGIS BANK

Database Standards

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This document defines database architecture, security controls, operational procedures, and management standards used by Aegis Bank.

The purpose of these standards is to ensure secure, reliable, scalable, and compliant management of banking data.

Unauthorized access, modification, or distribution of database information is strictly prohibited.

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1. Purpose

The purpose of this document is to establish database management standards for Aegis Bank systems.

These standards ensure:

- Secure storage of banking information.
- High availability of critical systems.
- Data accuracy and integrity.

- Regulatory compliance.
- Efficient database operations.

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2. Scope

This policy applies to:

- Database administrators.
- Backend developers.
- Data engineers.
- DevOps teams.
- Security teams.
- AI engineering teams.

It covers databases used for:

- Core banking systems.
- Customer management.
- Transactions.
- Loan processing.
- Reporting systems.
- AI knowledge platforms.

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3. Database Management Principles

Aegis Bank follows these principles:

Security First

All databases must implement security controls from design to operation.

Data Integrity

Data must remain:

- Accurate.
- Consistent.
- Reliable.

Availability

Critical banking databases must maintain high availability.

Scalability

Database systems must support future business growth.

Compliance

Database operations must follow:

- Privacy policies.
- Security standards.
- Regulatory requirements.

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4. Database Architecture Standards

Aegis Bank uses a layered database architecture.

Application Layer

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API Services

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Database Layer

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Storage Infrastructure

Database Environments

All databases must have separate environments:

Development

Used for:

- Coding.
- Local testing.
- Feature development.

Testing

Used for:

- Integration testing.
- Quality validation.

Staging

Used for:

- Production simulation.
- Final approval.

Production

Used for:

- Live banking operations.

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5. Database Technology Standards

Aegis Bank supports approved database technologies.

Relational Databases

Used for structured banking data.

Examples:

- PostgreSQL.
- Microsoft SQL Server.
- Oracle Database.

Used for:

- Customer accounts.
- Transactions.
- Financial records.

NoSQL Databases

Used for flexible and high-volume data.

Examples:

- MongoDB.

Used for:

- Application data.
- Logs.
- Non-structured information.

Vector Databases

Used for AI and RAG systems.

Examples:

- Pinecone.
- FAISS.
- ChromaDB.

Used for:

- Document embeddings.
- Semantic search.
- Knowledge retrieval.

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6. Data Modeling Guidelines

Database designs must follow standard modeling practices.

Requirements:

- Clear naming conventions.
- Proper relationships.
- Data normalization.
- Documentation.

Table Naming Standards

Correct:

customer_accounts

loan_applications

transaction_history

Incorrect:

table1

data_new

customer123

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7. Database Security Requirements

All databases must implement:

- Authentication.
- Authorization.
- Encryption.
- Monitoring.
- Audit logging.

Security Controls

Required controls:

- Strong passwords.
- MFA for administrators.
- Network restrictions.
- Firewall protection.
- Regular security reviews.

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8. Access Control Standards

Database access follows:

Principle of Least Privilege

Users receive only required permissions.

Role-Based Access Control (RBAC)

Example:

Role	Access
Developer	Development database
Analyst	Approved reports
DBA	Database administration
Security Team	Audit monitoring

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9. Encryption Requirements

Aegis Bank protects data using encryption.

Data Encryption at Rest

Protects stored information.

Examples:

- Customer records.
- Transaction data.
- Backup files.

Data Encryption in Transit

Protects information moving between systems.

Requirements:

- TLS encryption.
- Secure connections.

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10. Backup & Recovery Standards

Critical databases must have regular backups.

Backup types:

Full Backup

Complete database copy.

Incremental Backup

Only changed data is stored.

Backup Requirements

Backups must be:

- Encrypted.
- Tested.
- Monitored.
- Stored securely.

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11. Database Performance Management

Database performance is monitored continuously.

Metrics include:

- Query response time.
- CPU usage.
- Memory usage.
- Connection count.
- Storage utilization.

Optimization methods:

- Query optimization.
- Index management.
- Database tuning.

- Resource scaling.

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12. Data Integrity Controls

Aegis Bank maintains data accuracy through:

- Validation rules.
- Constraints.
- Transaction controls.
- Audit trails.

Examples:

A transaction cannot be completed if:

- Account does not exist.
- Required approval is missing.
- Security verification fails.

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13. AI & Vector Database Standards

Aegis AI OS uses vector databases for RAG-based knowledge retrieval.

Vector Database Requirements

Must support:

- Secure document storage.
- Embedding management.
- Semantic search.
- Access control.

AI Knowledge Flow

Enterprise Documents

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Document Processing

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Text Embeddings

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Vector Database

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Semantic Search

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LLM Response

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14. Database Monitoring

Database systems must be continuously monitored.

Monitoring includes:

- Performance.

- Security events.
- Failed access attempts.
- Database availability.
- Backup status.

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15. Disaster Recovery Requirements

Critical databases must support recovery procedures.

Requirements:

- Backup restoration testing.
- Replication.
- Failover planning.
- Recovery documentation.

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16. Database Security Architecture

Aegis Bank follows a multi-layer database security architecture to protect sensitive banking information.

Application Users

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API Security Layer

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Authentication Service

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Authorization Controls

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Database Firewall

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Encrypted Database

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Secure Storage

Database Security Controls

Authentication

All database users must authenticate using approved methods.

Requirements:

- Strong credentials.
- Multi-Factor Authentication for administrators.
- Secure service accounts.
- Credential rotation.

Authorization

Database permissions must be assigned according to job responsibilities.

Example:

User Type	Permission
Application Service	Read/Write required tables
Developer	Development database only
Database Administrator	Full database management
Auditor	Read-only access

Database Firewall

Database access must be restricted through:

- Network rules.
- IP restrictions.
- Private connections.
- Access monitoring.

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17. Database Change Management

All database changes must follow a controlled approval process.

Changes include:

- Creating new tables.
- Modifying database structure.
- Updating indexes.
- Changing permissions.

Change Workflow

Change Request

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Impact Analysis

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Testing Environment

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Security Review

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Approval

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Production Deployment

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18. Database Audit Logging

Aegis Bank maintains database activity records.

Logs include:

- User access.
- Data modifications.

- Failed login attempts.
- Administrative activities.
- Permission changes.

Audit Log Protection

Database logs must be:

- Securely stored.
- Protected from modification.
- Regularly reviewed.

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19. Data Privacy Controls

Customer information stored in databases must follow privacy requirements.

Controls include:

- Data masking.
- Access restrictions.
- Encryption.
- Data classification.

Example: Customer Account Data

Restricted:

Account Number

Transaction History

Identity Information

Protection:

- Masking in reports.
- Limited employee access.

- Audit tracking.

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20. Database Backup Strategy

Aegis Bank follows a structured backup strategy.

Backup Schedule

Backup Type	Frequency
Incremental Backup	Daily
Full Backup	Weekly
Disaster Recovery Backup	According to business requirements

Backup Security

Backups must:

- Be encrypted.
- Have restricted access.
- Be regularly tested.
- Maintain integrity checks.

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21. Database Recovery Process

When database failure occurs:

Incident Detection

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System Assessment

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Recovery Decision

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Backup Restoration

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Data Validation

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Service Recovery

Recovery Validation

After restoration:

- Data accuracy is verified.
- Applications are tested.
- Security controls are checked.

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22. High Availability Database Standards

Critical banking databases must support high availability.

Methods include:

- Database replication.
- Failover systems.
- Load balancing.
- Cluster architecture.

Availability Requirements

Systems must minimize:

- Downtime.
- Data loss.
- Service interruptions.

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23. RAG Knowledge Database Standards

Aegis AI OS requires specialized database standards for enterprise knowledge retrieval.

Document Processing Pipeline

PDF Documents

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Text Extraction

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Cleaning & Processing

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Embedding Generation

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Vector Database Storage

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Similarity Search

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LLM Response Generation

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24. Vector Database Security

Vector databases containing enterprise knowledge must implement:

- Authentication.
- Access control.
- Encryption.
- Document permissions.

AI Retrieval Rules

Aegis AI OS must:

- Retrieve only authorized documents.
- Respect department permissions.

- Maintain retrieval logs.
- Track source references.

Example:

Employee asks:

"What is the password policy?"

AI Process:

1. Verify employee access.
2. Search Security documents.
3. Retrieve AB-SEC-001.
4. Generate response.
5. Show source reference.

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25. Database Performance Optimization

Database teams continuously improve performance through:

Query Optimization

Activities:

- Reviewing slow queries.
- Improving SQL statements.
- Reducing unnecessary operations.

Index Management

Indexes improve:

- Search speed.
- Query performance.
- Data retrieval efficiency.

Resource Management

Monitoring:

- CPU.
- Memory.
- Storage.
- Network usage.

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26. Database Responsibilities

Chief Technology Officer (CTO)

Responsible for:

- Database strategy.
- Technology decisions.
- Architecture approval.

Database Administrators (DBA)

Responsible for:

- Database maintenance.
- Security configuration.
- Backup management.
- Performance optimization.

Developers

Responsible for:

- Writing efficient queries.
- Following database standards.

- Protecting sensitive data.

Security Team

Responsible for:

- Database security reviews.
- Access monitoring.
- Compliance checks.

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27. Practical Banking Database Scenarios

Scenario 1 — Unauthorized Database Access

Situation:

An employee attempts to access customer transaction records without permission.

Response:

- Access blocked.
- Event logged.
- Security team notified.
- Investigation started.

Scenario 2 — Database Server Failure

Situation:

Production database becomes unavailable.

Response:

1. Incident detected.
2. Failover system activated.
3. Database restored.

4. Services validated.

Scenario 3 — AI Knowledge Retrieval

Situation:

Employee asks:

"What is Aegis Bank's AML policy?"

Process:

1. Query sent to AI system.
2. Vector database searched.
3. Relevant document retrieved.
4. Source displayed.
5. Answer generated.

Scenario 4 — Backup Restoration

Situation:

Database corruption occurs.

Response:

- Latest verified backup selected.
- Data restored.
- Integrity checked.
- Service resumed.

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28. Frequently Asked Questions (FAQs)

Q1. Why are database standards important?

They ensure banking data remains secure, reliable, and compliant.

Q2. Who can access production databases?

Only authorized personnel with approved permissions.

Q3. How does Aegis Bank protect customer data?

Through encryption, access control, monitoring, and security policies.

Q4. How does AI use database systems?

Aegis AI OS uses vector databases to securely retrieve enterprise documents for RAG responses.

Q5. How often are backups performed?

Backup frequency depends on system importance and business requirements.

Q6. What happens during database failure?

Recovery procedures restore systems using approved backup and failover processes.

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29. Related Documents

This document should be read with:

- AB-IT-001 — API Development Standards
- AB-IT-003 — Docker Deployment Guide
- AB-IT-004 — CI/CD Standards
- AB-IT-005 — Logging & Monitoring Standards
- AB-SEC-003 — Cyber Security Policy
- AB-COMP-006 — Customer Data Retention Policy

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30. Revision History

Version	Date	Description	Author
v1.0	March 12, 2026	Initial Database Standards Release	Database Engineering Team
v1.1	May 30, 2026	Added Security and AI Database Controls	IT Architecture Team
v1.2	**July 24, 2026**	**Enhanced Database Security, RAG Infrastructure, Backup, and Recovery Standards**	**Information Technology Department**

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31. Document Approval

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End of Document

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